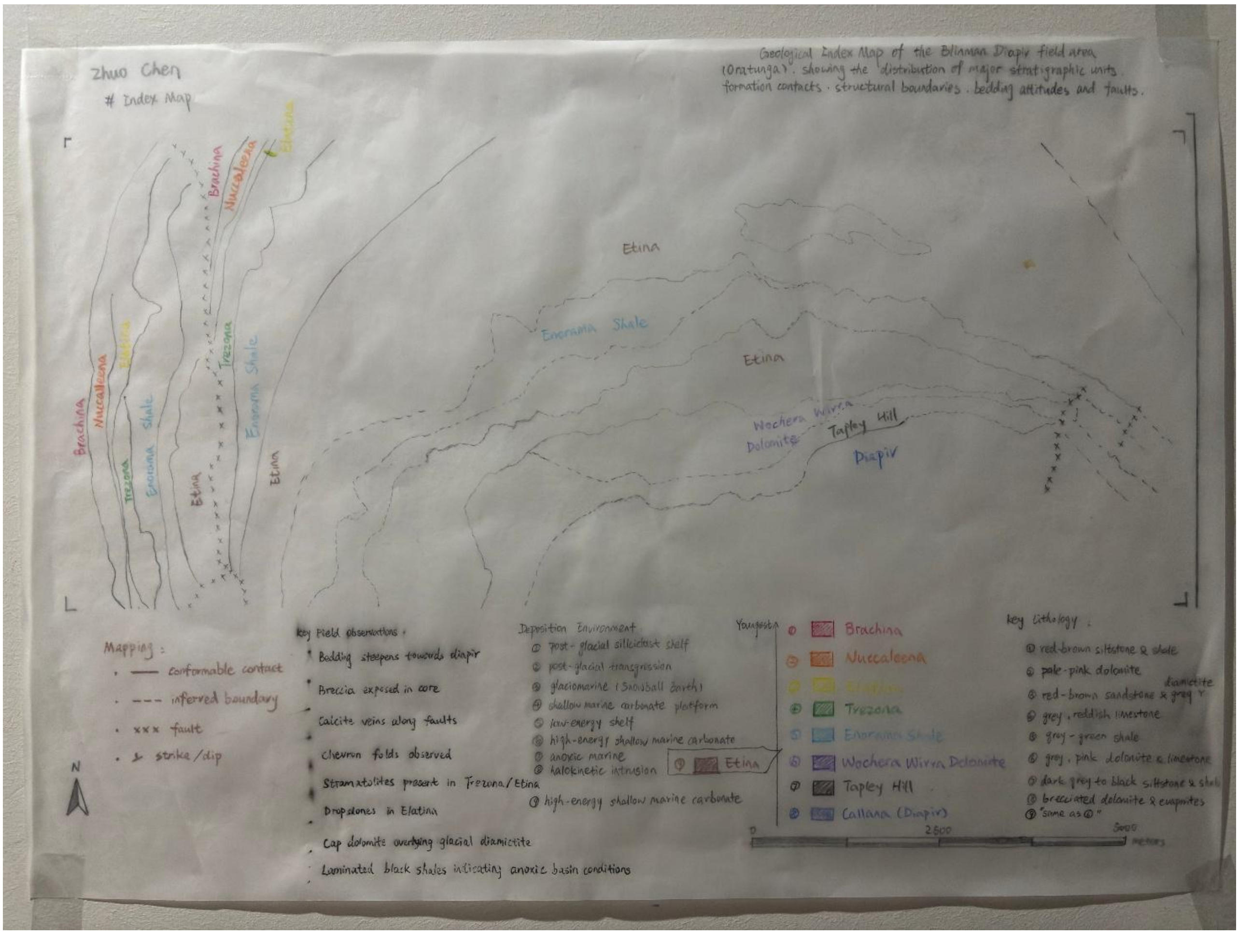
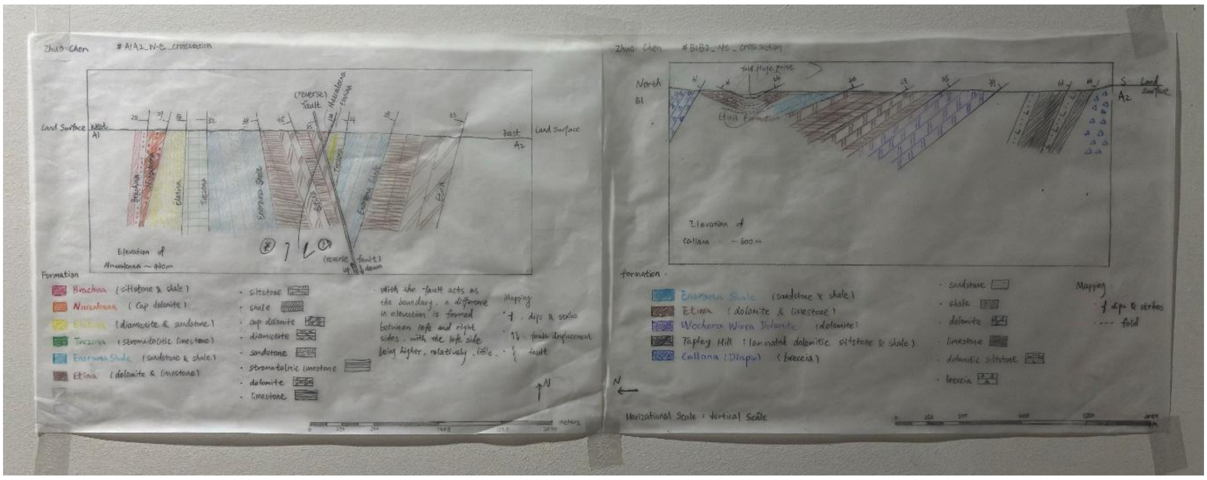
在偏远且构造复杂的地区（澳大利亚阿德莱德）开展为期一周的团队野外作业，采集并整理 60 余组构造测量数据，保持观察和测量记录的准确性与可追溯性。使用 Avenza Maps 进行现场导航、空间定位及数据采集，使用 ArcGIS 整理和可视化现场信息。与组员沟通数据差异和解释结果，在遵守安全与环境要求的前提下完成现场任务，积累复杂环境下的执行、核对和协作经验。最终完成 6 张工作图的资料汇编与交叉核对，整合不同来源的观察和测量结果，形成完整地质图及 2 条地质剖面。

完整地质图：

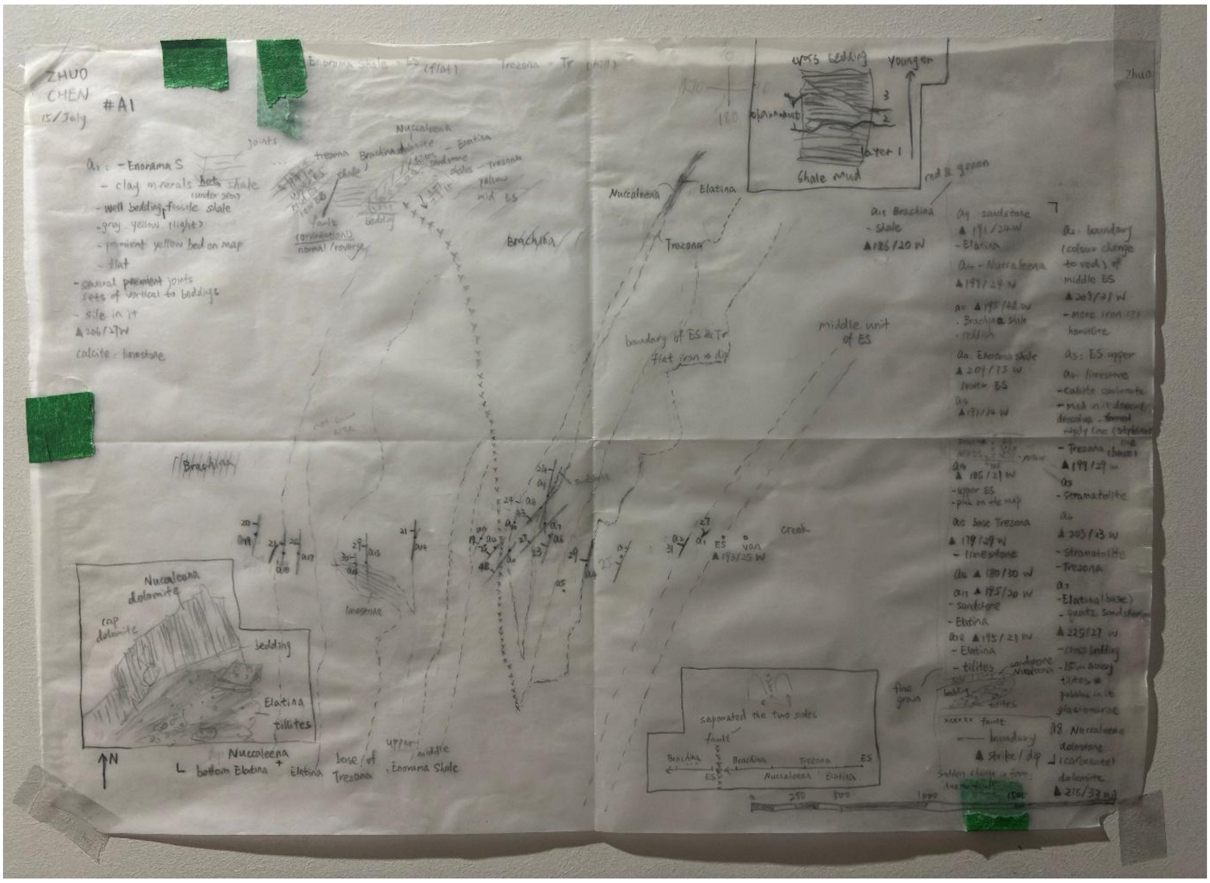

Zhuo Chen
# Index Map
Geological Index Map of the Blinman Diapir field area (Oratunga), showing the distribution of major stratigraphic units, formation contacts, structural boundaries, bedding attitudes and faults.
Brachina
Nuccaleena
Elatina
Trezona
Enorama Shale
Etina
Etina
Enorama Shale
Etina
Wochera Wirra Dolomite
Tapley Hill
Diapir
Mapping:
conformable contact
inferred boundary
xxx fault
strike/dip
N
Key Field observations:
Bedding steepens towards diapir
Breccia exposed in core
Calcite veins along faults
Chevron folds observed
Stromatolites present in Trezona/Etina
Dropstones in Elatina
Cap dolomite overlying glacial diamictite
Laminated black shales indicating anoxic basin conditions
Deposition Environment
① post-glacial siliciclastic shelf
② post-glacial transgression
③ glaciomarine (Snowball Earth)
④ shallow marine carbonate platform
⑤ low-energy shelf
⑥ high-energy shallow marine carbonate
⑦ anoxic marine
⑧ halokinetic intrusion
⑨ high-energy shallow marine carbonate
Youngest
① Brachina
② Nuccaleena
③ Elatina
④ Trezona
⑤ Enorama Shale
⑥ Wochera Wirra Dolomite
⑦ Tapley Hill
⑧ Callana (Diapir)
⑨ Etina
Key Lithology:
① red-brown siltstone & shale
② pale-pink dolomite
③ red-brown sandstone & grey diamictite
④ grey, reddish limestone
⑤ grey-green shale
⑥ grey, pink dolomite & limestone
⑦ dark grey to black siltstone & shale
⑧ brecciated dolomite & evaporites
⑨ "same as ⑥"
0
2500
5000
meters

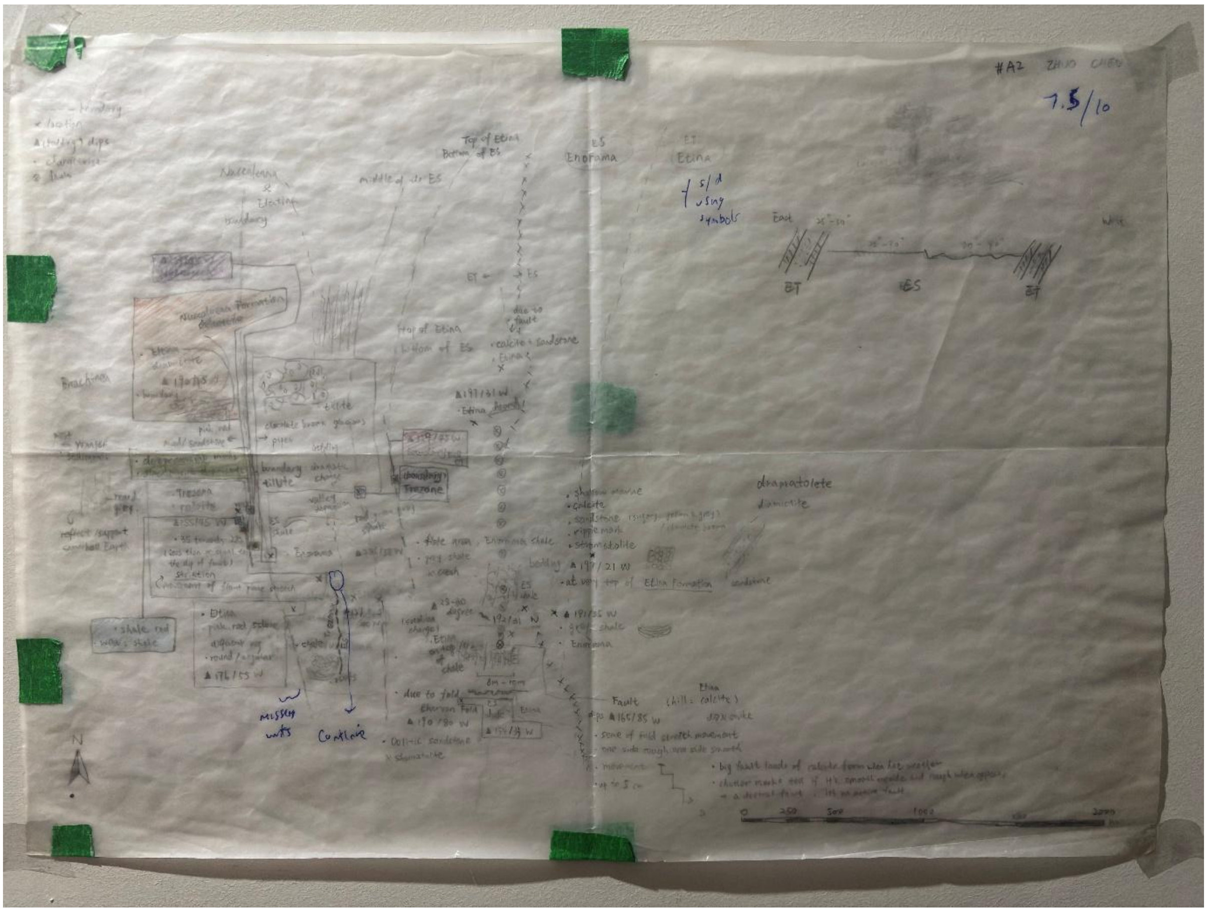
2 条地质剖面：

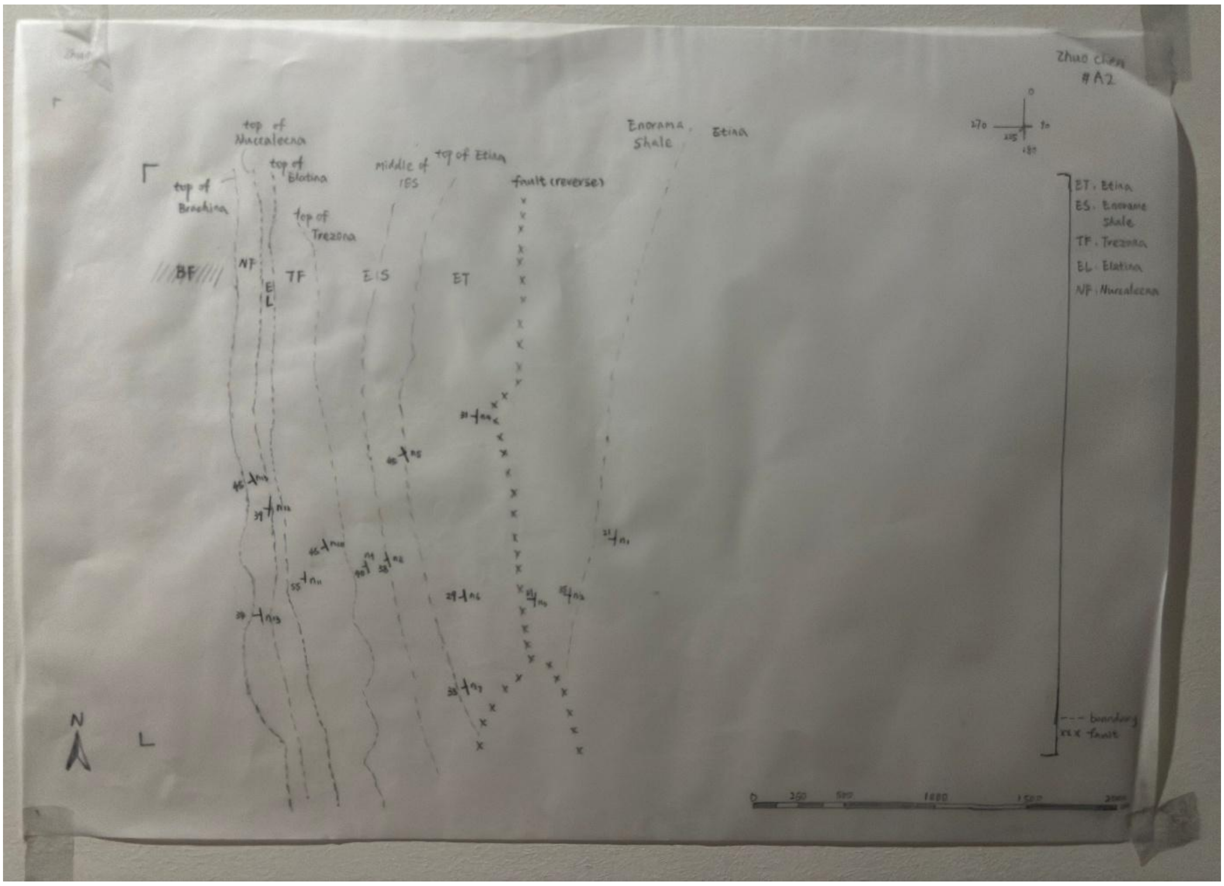

Zhuo Chen
# A1A2_W-E_crosssection
Land Surface
West
A1
East
A2
Land Surface
(reverse) fault
Nuccaleena
Elatina
Brachina
Trezona
Enorama Shale
Etina
Elevation of Nuccaleena ~ 430m
(reverse fault)
up
down
Formation
Brachina (siltstone & shale)
Nuccaleena (Cap dolomite)
Elatina (diamictite & sandstone)
Trezona (stromatolitic limestone)
Enorama Shale (sandstone & shale)
Etina (dolomite & limestone)
· siltstone
· shale
· cap dolomite
· diamictite
· sandstone
· stromatolitic limestone
· dolomite
· limestone
With the fault acts as the boundary, a difference in elevation is formed between left and right sides, with the left side being higher, relatively little.
Mapping
· dips & strike
· fault displacement
· fault
N
meters
Zhuo Chen
#B1B2_NS_cross section
fold hinge point
North
B1
Etina formation
S
A2
Land Surface
Elevation of Callana ~ 600 m
formation :
Enorama Shale (sandstone & shale)
Etina (dolomite & limestone)
Wocheera Wirra Dolomite (dolomite)
Tapley Hill (laminated dolomitic siltstone & shale)
Callana (Diapir) (breccia)
· sandstone
· shale
· dolomite
· limestone
· dolomitic siltstone
· breccia
Mapping
dip & strikes
fold
N
Horizational Scale : Vertical Scale

6 张工作图：

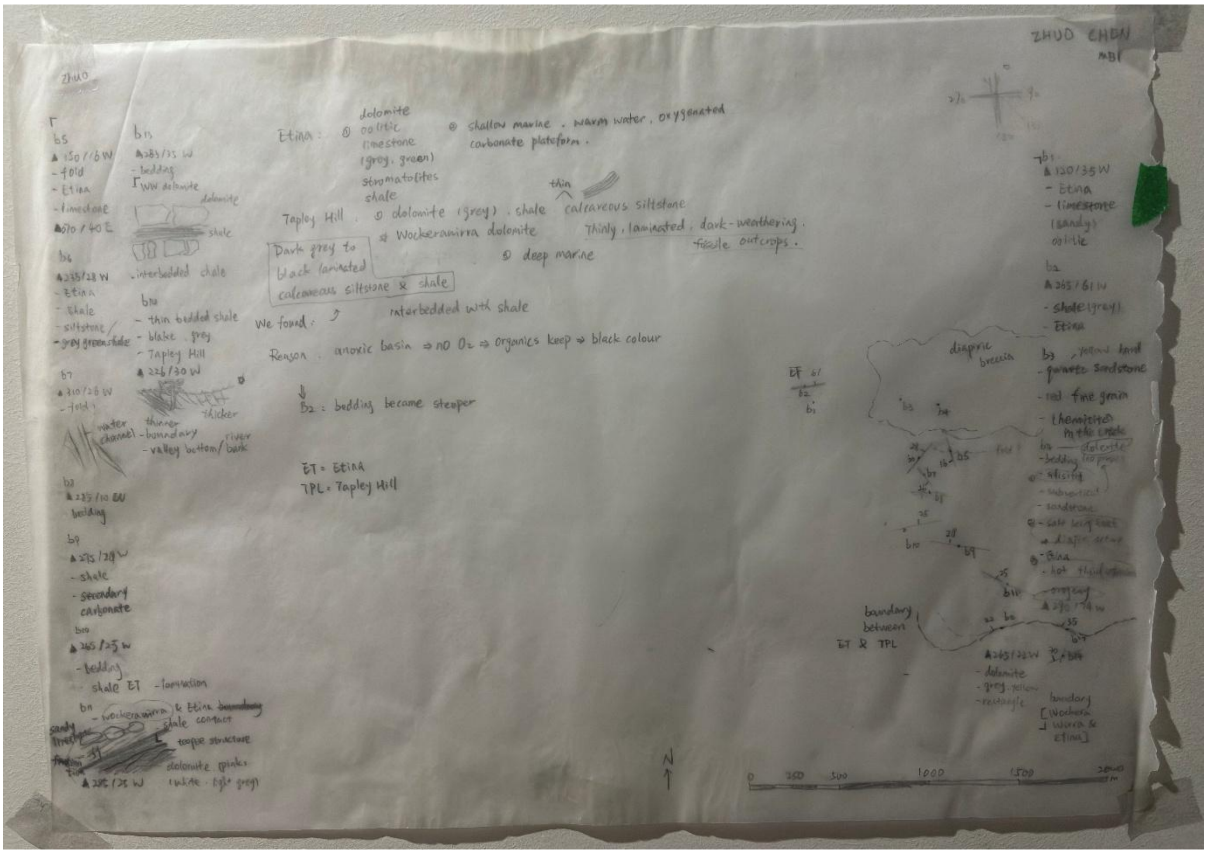
- A 区

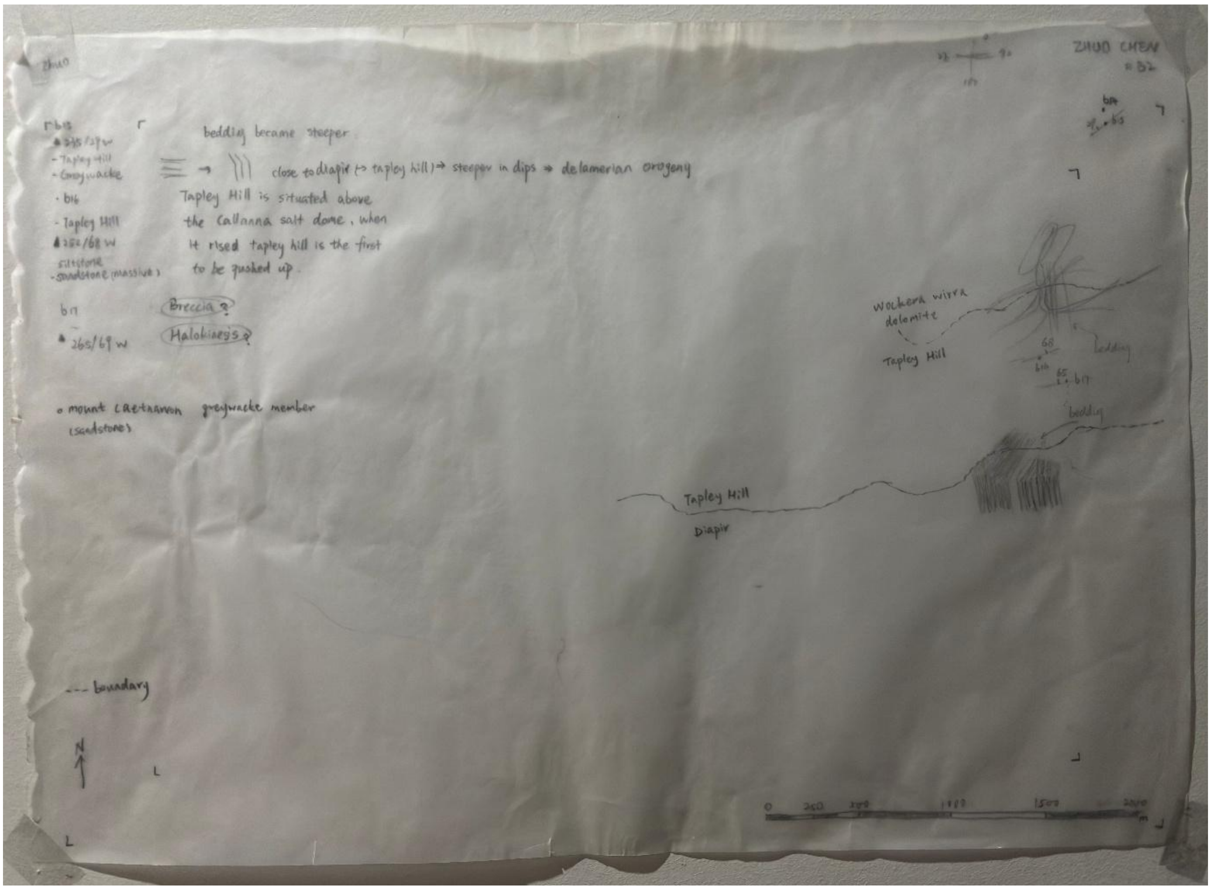



#A2 ZHUO CHEN
7.5/10
boundary
location
bedding + dips
Nuccaleena & Elatina boundary
middle of the ES
Top of Etina Bottom of ES
ES Enorama
ET Etina
s/d using symbols
Nuccaleena Formation
Brachina
Etina diamictite
▲170/85 W
boundary
pink, red mud/sandstone
younger sediments
tillite
chocolate brown grooves
pipes
bedding
boundary tillite
dramatic change
boundary Trezona
Trezona
calcite
reflect/support snowball Earth
▲155/95 W
striation
valley
ES shale
Enorama
ET ← → ES
due to fault
top of Etina bottom of ES
calcite + sandstone
▲197/31 W
Etina
flate area, Enorama shale
grey shale
shale red
Etina
pink, red, yellow
round/angular
▲176/55 W
missing units
Continue
N
25-80 degree
192/31 W
due to fold
Chevron Fold
▲170/80 W
Oolitic sandstone
stromatolite
8m - 10m
▲154/13 W
shallow marine
calcite
sandstone
ripple mark
stromatolite
bedding ▲197/21 W
at very top of Etina Formation
sandstone
▲191/35 W
grey shale
Enorama
Fault
dips ▲165/85 W
Etina (hill: calcite)
diamictite
East
West
ET
ES
ET
0
250
500
1000
2000

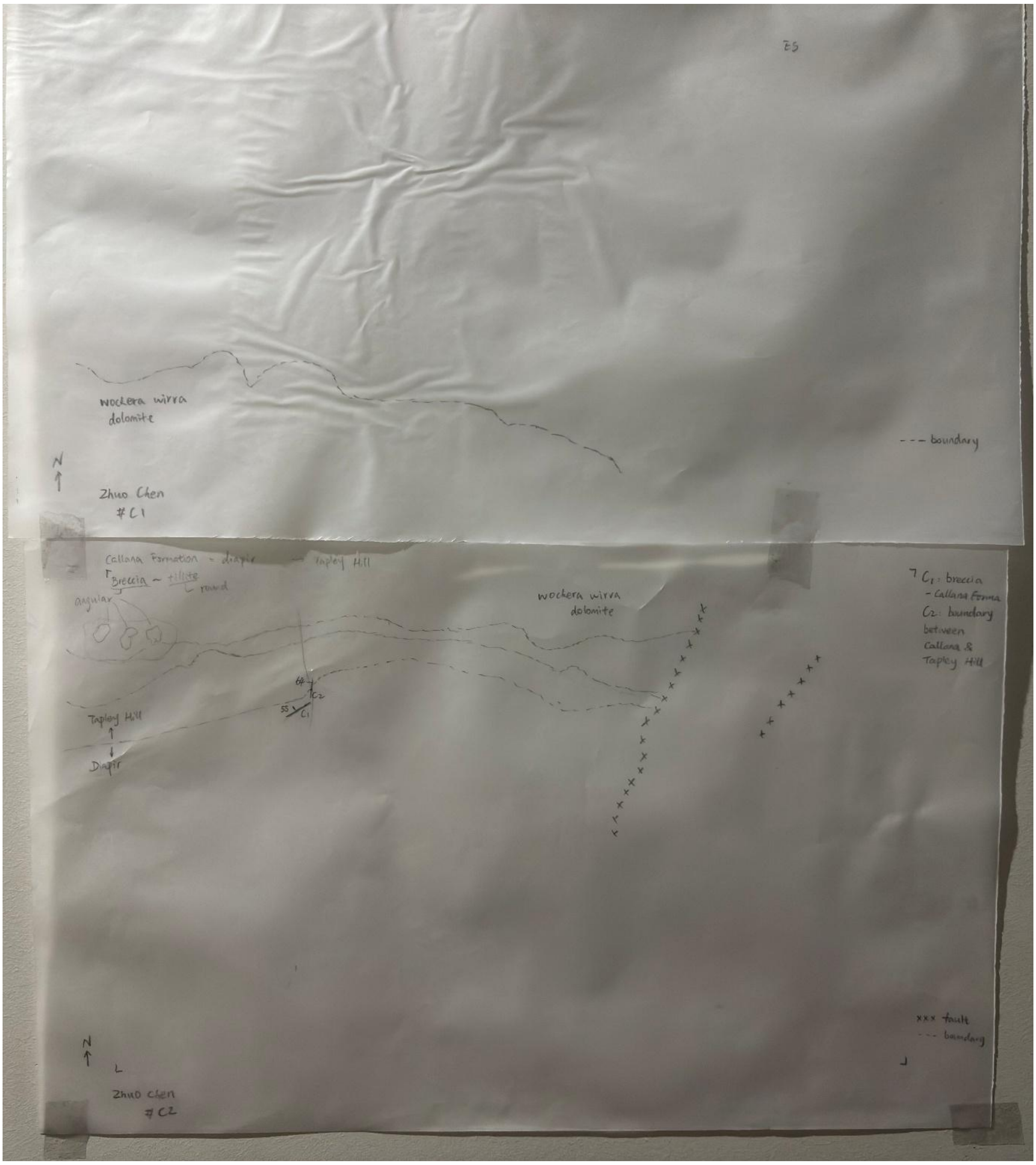

Zhuo Chen
#A2
0
90
180
270
top of Nuccaleena
top of Elatina
top of Brachina
top of Trezona
middle of ES
top of Etina
fault (reverse)
Enorama Shale
Etina
BF
NF
EL
TF
ES
ET
ET: Etina
ES: Enorama Shale
TF: Trezona
EL: Elatina
NF: Nuccaleena
boundary
fault
N
0
250
500
1000
1500
2000

- B 区


ZHUO
ZHUO CHEN
MB1
b5
▲150/16W
- fold
- Etina
- limestone
▲570/40E
b6
▲235/28W
- Etina
- shale
- siltstone/
- grey greenshale
b7
▲310/26W
- fold
water channel
b13
▲283/35W
- bedding
WW dolomite
dolomite
shale
- interbedded shale
b14
- thin bedded shale
- blake grey
- Tapley Hill
▲226/30W
thinner
thicker
- boundary
- valley bottom/river bank
b8
▲225/10W
bedding
b9
▲275/29W
- shale
- secondary carbonate
b10
▲265/25W
- bedding
shale ET - formation
b11
- wockeranirra & Etina ~~boundary~~
shale contact
teepee structure
dolomite (pink white light grey)
▲205/25W
Etina: ① dolomite oolitic limestone (grey, green) stromatolites shale
② shallow marine, warm water, oxygenated carbonate plateform.
Tapley Hill: ① dolomite (grey), shale, thin calcareous siltstone
② Wockeranirra dolomite
Thinly, laminated, dark-weathering, fissile outcrops.
③ deep marine
Dark grey to black laminated calcareous siltstone & shale
We found: interbedded with shale
Reason: anoxic basin ⇒ no O2 ⇒ organics keep ⇒ black colour
B2: bedding became steeper
ET = Etina
TPL = Tapley Hill
ET 61
b2
b1
diapiric breccia
b3
b4
b5
b7
b8
b10
b9
b12
b6
b17
fold
boundary between ET & TPL
b1
▲130/35W
- Etina
- limestone (sandy) oolitic
b2
▲265/61W
- shale (grey)
- Etina
b3, yellow band
- quartz sandstone
- red fine grain
- hematite in the crack
b4 — dolerite
- bedding
▲265/22W
- dolomite
- grey yellow
- rectangle
boundary [Wockera wirra & Etina]
N
0 250 500 1000 1500 2000 m


ZHUO
ZHUO CHEN
B2
b15
▲245/29W
- Tapley Hill
- Greywacke
· b16
- Tapley Hill
▲250/68W
- siltstone sandstone (massive)
b17
▲265/69W
bedding became steeper
close to diapir (→ tapley hill) → steeper in dips ⇒ delamerian orogeny
Tapley Hill is situated above the Callanna salt dome, when it rised tapley hill is the first to be pushed up.
Breccia?
Halokinesis?
· mount caernarvon greywacke member (sandstone)
--- boundary
N
b14
b15
Wockera wirra dolomite
Tapley Hill
68
b16
65
b17
bedding
Tapley Hill
Diapir
0 250 500 1000 1500 2000 m

- C 区


E5
wockera wirra
dolomite
--- boundary
N
Zhuo Chen
#C1
Callana Formation = diapir ~ Tapley Hill
Breccia ~ tillite
round
angular
wockera wirra
dolomite
64
C2
55
C1
Tapley Hill
Diapir
C1: breccia
- Callana Forma
C2: boundary
between
Callana &
Tapley Hill
xxx fault
--- boundary
N
Zhuo Chen
#C2